Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/13/2023
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Does Intermittent or Continuous Artificial Gravity Counteract Long Duration Bed Rest Induced Neurocognitive Declines?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rachaelseidler@ufl.edu	**Fax:**	FY
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Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-AGBR. Appendix G: Physiological & Behavioral Responses in Humans to Intermittent Artificial Gravity during Bed Rest
Start Date:	07/01/2018	**End Date:**	12/31/2022
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2022 per L. Juliette/JSC (Ed., 6/2/22). NOTE: End date changed to 6/30/2022 per NSSC information (Ed., 6/18/21) NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 8/28/20)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Mulavara, Ajitkumar Ph.D. (KBR/NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Bloomberg, Jacob Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC18K0783		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Research Announcement NNJ15ZSA001N-AGBR: Appendix G: Physiological and Behavioral Responses in Humans to Intermittent Artificial Gravity during Bed Rest invites proposals to investigate whether 30 minutes of artificial gravity (AG) per day, delivered either continuously or in six, five minute intermittent blocks, mitigates changes that occur in a spaceflight analog environment. We have recently completed a 70 day 6° head down tilt bed rest study in which we identified specific bed rest induced changes in brain structure, brain function, sensorimotor, and cognitive performance. Here, we select those that are sensitive to this spaceflight analog environment as outcome metrics for the following aims: Aim 1) Determine whether 30 minutes per day of supine centrifugation +1Gz at the heart is effective at mitigating neurocognitive declines induced by 60 days of 6° head down tilt bed rest in comparison to a control group that is not exposed to AG. Aim 2) Determine whether centrifugation is more effective and better tolerated as a countermeasure when it is delivered continuously for 30 minutes or in six, five minute sessions separated by breaks of five minutes each. Our approach utilizes cutting edge neuroimaging techniques and a broad ranging battery of sensory, motor, and cognitive assessments that we have found to be sensitive indices of bed rest—and in several cases—spaceflight induced performance decrements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research on adaptive plasticity associated with behavioral training has greatly expanded over the past two decades, but research on maladaptive plasticity occurring with immobilization is scant. A greater understanding of brain structural and functional changes, and the concomitant behavioral effects resulting from limb disuse and unloading has implications for rehabilitation of those immobilized by injury, disease, or even simple inactivity.
Task Progress:	This project was conducted as a part of the “AGBRESA” campaign at envihab in Cologne, Germany. This was a 60-day head down tilt bed rest study with daily centrifugation as a potential countermeasure. One group did not participate in centrifugation, one received 30 minutes per day of continuous centrifugation, and one received 30 minutes per day of intermittent centrifugation in 5-minute sessions. The original aims of the project are: Aim 1) Determine whether 30 minutes per day of supine centrifugation +1Gz at the heart is effective at mitigating declines in brain structure, brain function, sensorimotor and cognitive performance induced by 60 days of 6° head down tilt bed rest in comparison to a control group that is not exposed to AG. Aim 2) Determine whether centrifugation is more effective and better tolerated as a countermeasure when it is delivered continuously for 30 minutes or in six, five minute sessions separated by breaks of five minutes each. Aim 3) Identify predictors of individual differences in bed rest and AG responses. We will examine whether pre bed rest measures of brain structure and brain function are predictive of changes in sensory, motor, and cognitive function occurring in association with bed rest, and whether they also predict AG tolerance. We will combine data collected here with those being acquired for two funded “predictors” studies (Bloomberg, Mulavara and Wood, PIs) to increase our predictive power. The data were collected at :envihab, with data collection complete in December, 2019. Data transfer from Germany to the Seidler laboratory was completed in March, 2020. We had an initial slowing of data analyses due to the COVID pandemic. We are wrapping up a few final analyses and manuscript submissions, which will be submitted over the coming few months. In summary, our results reveal that daily centrifugation mitigates some of the behavioral changes that occur with long duration head down tilt bed rest. We also found that the intermittent artificial gravity group was less susceptible to post-rotation illusory motion. In some cases, effect sizes were moderate, but the results were not statistically significant. This could be due to the small sample sizes per group. It could also be because several of our measures rely on vestibular inputs, and the centrifugation only amounted to ~0.3 g at the level of the head. We did not find that centrifugation reduced intracranial fluid shifts, as quantified via free water analyses. We did find several cases of group x time differences in brain functional activity, though, during the performance of a range of tasks. These changes largely suggest greater neural efficiency in the artificial gravity groups, and / or differential sensory reweighting between the artificial gravity and control groups.
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Articles in Peer-reviewed Journals	Tays GD, Hupfeld KE, McGregor HR, Gadd NE, Kofman IS, De Dios YE, Mulder ER, Bloomberg JJ, Mulavara AP, Wood SJ, Seidler RD. "Daily artificial gravity is associated with greater neural efficiency during sensorimotor adaptation." Cereb Cortex. 2023 Jun 8;33(12):8011-23. https://doi.org/10.1093/cercor/bhad094 ; PMID: 36958815; PMCID: PMC10267627 , Jun-2023
Articles in Peer-reviewed Journals	McGregor HR, Lee JK, Mulder ER, De Dios YE, Beltran NE, Wood SJ, Bloomberg JJ, Mulavara AP, Seidler RD. "Artificial gravity during a spaceflight analog alters brain sensory connectivity." Neuroimage. 2023 Sep;278:120261. https://doi.org/10.1016/j.neuroimage.2023.120261 ; PMID: 37422277 , Sep-2023
Articles in Peer-reviewed Journals	Tays GD, Hupfeld KE, McGregor HR, Beltran NE, De Dios YE, Mulder E, Bloomberg JJ, Mulavara AP, Wood SJ, Seidler RD. "Daily artificial gravity partially mitigates vestibular processing changes associated with head-down tilt bedrest." npj Microgravity. 2024 Mar 12;10:27.https://doi.org/10.1038/s41526-024-00367-7 ; PMID: 38472244; PMCID: PMC10933323 , Mar-2024
Articles in Peer-reviewed Journals	Tays GD, Hupfeld KE, McGregor HR, Banker LA, De Dios YE, Bloomberg JJ, Reuter-Lorenz PA, Mulavara AP, Wood SJ, Seidler RD. "The microgravity environment affects sensorimotor adaptation and its neural correlates." Cereb Cortex. 2025 Feb 5;35(2):bhae502. https://doi.org/10.1093/cercor/bhae502 ; PMID: 39756418; PMCID: PMC11795311 , Feb-2025
Articles in Peer-reviewed Journals	Richmond SB, Seidler RD, Iliff JJ, Schwartz DL, Luther M, Silbert LC, Wood SJ, Bloomberg JJ, Mulder E, Lee JK, De Luca A, Piantino J. "Dynamic changes in perivascular space morphology predict signs of spaceflight-associated neuro-ocular syndrome in bedrest." npj Microgravity. 2024 Mar 1;10:24.https://doi.org/10.1038/s41526-024-00368-6 ; PMID: 38429289; PMCID: PMC10907584 , Mar-2024
Articles in Peer-reviewed Journals	Tays GD, McGregor HR, De Dios YE, Mulder E, Bloomberg JJ, Mulavara AP, Wood SJ, Seidler RD. "Thirty minutes of daily artificial gravity does not mitigate head down tilt induced brain activity changes during cognitive task performance." Front Neurol. 2025 Aug 26;16:1602104. https://doi.org/10.3389/fneur.2025.1602104 ; PMID: 40937179; PMCID: PMC12421903 , Aug-2025

Articles in Peer-reviewed Journals	Wang T, Odor RJ, Dios YE, Mulder E, Bloomberg JJ, Wood SJ, Seidler RD. "Brain displacement and nonlinear deformation following human spaceflight." Proc Natl Acad Sci USA. 2026 Jan 20;123(3):e2505682122. https://doi.org/10.1073/pnas.2505682122 ; PMID: 41525486 , Jan-2026